Forex Market Part 2

Financial Institutions and Markets

Readings

http://www.economist.com/blogs/economist-explains/2013/09/economist-explains-1

https://www.youtube.com/watch?v=2xIgPYcN288

Topics

- Reasons for and effects of forex volatility
- Forms of exposure to forex risk
- Difference between spot and forward rates
- How to hedge forex risk
- A proposed solution

Discussion

- Imagine you have an idea for a new business
  - You will bottle "mia quat" and sell it in the US
- Your workers, land, machines, etc. are all paid in VND

You have to make long-term commitments

- Multi-year marketing contracts (in USD)
- Buying large amounts of land (with loans in VND)
- Multi-year labor contracts (in VND)

- Are you worried about changes in the USD:VND rate?
- Could this make you skip this opportunity?

Problems from forex variability

- Uncertainty about future export revenues
- Uncertainty about future import costs
- Increased costs from the need to engage in hedging
- More exchange rate volatility= greater these problems
- Volatility: frequency and magnitude variations
  - 1.30 to 1.31 to 1.30=low volatility; 1.30 to 1.45 to 1.25=high
- As the following data show, exchange rates can vary significantly over even a short space of time.

Change in monthly £:$ (2010)

Month	%-change
January	0.00
February	-3.7
March	-3.21
April	1.32
May	-4.58
June	1.37
July	3.38
August	2.61
September	-0.64
October	1.92
November	0.63
December	-2.5

- The rate difference between May and August here is 7.5% (9.25% between May and November).
- On $1m, that is $ 75,000 difference
- Imagine you lose that profit just because you time it wrong.

Exposure to forex risk

Transaction risk

- Current contracts (in foreign currency) lose value in home currency
  - Due to time between contract execution and settlement

Translation risk

- Assets (cash, buildings) priced in foreign currency lose value in home currency
  - Really an accounting issue

Economic risk

- Operating environment changes because of forex
  - Example: VND gets stronger so Chinese imports look cheaper.

Activity

If extreme exchange rate changes are bad for business, what can governments do to fix it?

Forex rate volatility

- More forex volatility → more difficult to predict future
  - Planning is much more difficult, errors become very expensive
- 1944-1972: Bretton Woods
  - Exchange rates kept within +/- 1% band (volatility was small)
  - All currencies pegged to the USD, USD pegged to gold
  - US had all the gold (after WWII)
- After 1972, rates floated
  - And volatility more extreme as a result.

Activity

Since Bretton Woods failed because of the US, what can the UK government do now to fix the problem of extreme exchange rate variation?

European exchange rate mechanism

- In the late 1970s, the ERM was created
  - Similar to Bretton Woods
- Required members to stabilize their currencies
  - By intervening in forex market, maintaining narrow band
- In 1992, ERM broke down
  - Similar to Bretton Woods
  - Thanks to George Soros.

Activity

- UK exporter sells goods to French firm for €5,000,000. Payment due in three months.
  - Ex-rate: €1.1787:£
- Before payment, pound strengthens: €1.40:£
- How can UK firm protect itself?

Forward market

- At the time of contract, UK firms could ask their bank for a forward contract buying 5M worth of GBP at $1.1787
-
- If GBP strengthens to €1.40:£
  - The transaction loss (contract) = forward gain
- If GBP weakens to €1.10:£
  - The transaction gain (contract) = forward loss

Options market

Option: Contract giving the buyer the __right__ (but not the __obligation__) to a specific transaction.

- Call option: Right to __buy__ @ strike price
- Put option: Right to __sell__ @ strike price
- Options have an expiration date
- Option premium: Price of the right.

Spot vs forward rates

Interest rate parity theory:

- Difference between spot and forward markets = Difference between interest rates in two countries
- So, higher interest rates implies stronger currencies in the future.

Example

- UK gov't bonds: 6%; US gov't bonds: 8%
- Spot rate: $1.50:£, 1 year forward: $1.5283:£
- Choice A: Take £1m, buy UK debt
  - In one year: £1.06M (6%)
- Choice B: Take £1m, buy $1.5m, buy US debt
  - In one year: $1.62m (8%)
  - then trade $1.62m @ 1.5283 and get £ 1.06M.

Cause of exchange rate volatility

- Increasing exchange rate volatility has been linked to the growth in speculative currency transactions
- In the 1980s, daily forex = **$150 billion**
- By 2002, daily forex = **$1.8 trillion** (12x)
- Daily value of int'l trade in 2002 was just $90 billion
  - So only 5% could be related to trade financing (95%=speculation)
- Most currency transactions are of very short duration
- 80% of forex transactions are reversed within 7 days
  - 40% are reversed within 48 hours.

Problem and solution

- Exchange rate volatility is not only a problem for individual businesses
- Wild and unpredictable swings in exchange rates can have a destabilizing effect on national economies
  - Example: 1998 South-East Asia crisis
- The growth of speculative currency transactions has revived interest in a proposal for reducing exchange rate volatility from James Tobin in the 1970s.

Activity

If you want to reduce forex speculation, what could you do?

- Assume you are the government

James Tobin

- 1918-2002
- An American economist who was awarded the Nobel prize for economics in 1981
  - NOT for his _tax_ proposal
- He taught at Harvard and Yale universities
  - and was a neo-Keynesian (gov't should intervene in slowdown)
- He believed that speculation in foreign currencies was both damaging and _unproductive_.

The Tobin tax

Proposed by James Tobin in 1972

- A tax of _0.1_% to _0.5_% on all _spot_ forex transactions
- Such a low tax would reduce exchange rate volatility
  - without affecting normal business activity
- Even a low tax would lower speculation
  - since speculators tended to operate with very small _margins_
- Because each transaction gets taxed, even just 0.2%
  - daily speculators see an annual tax rate of about 50%
  - weekly would see 10%; monthly would see 2.4%.

Critique of the Tobin tax (1)

- Critics say, in practice it would do little to deter speculative currency transactions
- Even with a Tobin tax, small variations in exchange rates (just 3 or 4 per cent) would allow speculation
  - See chart next slide
- In practice, exchange rate variations often exceed 3 or 4 per cent
  - Like the 50% drop of the Thai baht, triggering the Asian crisis
- To be effective at reducing speculation, the Tobin tax would have to be set at such a high rate that it would damage normal business activity.

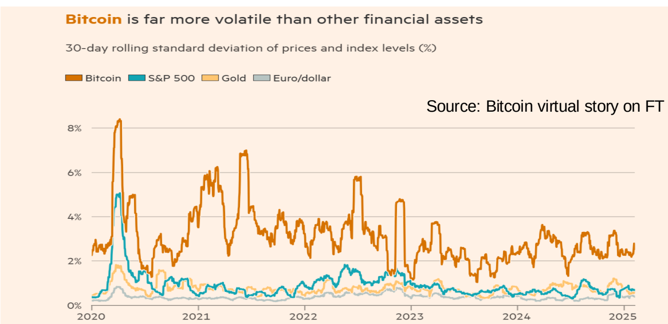
Multi-year volatility GBP:USD

Multi-day volatility GBP:USD

Critique of the Tobin tax (2)

- Maybe have a low rate for normal conditions, with a tax surcharge being applied at times of high volatility
  - To minimize the harmful impact on business
- In addition, a Tobin tax might be _costly_ to administer
- It might also encourage tax _avoidance_
  - people might use forward markets (this tax hits spot markets)
- It has little support from national governments
  - Although the idea has attracted considerable support from politicians, NGOs like Oxfam, and the academic community.

Prices for bitcoin and gold in US dollars



Source: Bitcoin virtual story on FT

What is crypto?

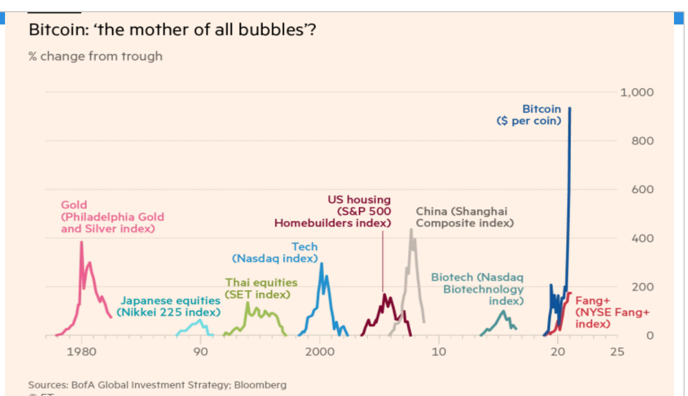
- Cryptocurrencies are defined as virtual currencies with a decentralised peer-to-peer electronic payment network without relying on a financial institution (Nakamoto, 2008)
- The literature on cryptocurrencies debate the definition of cryptocurrencies compared to fiat currencies. If cryptocurrencies can function as a medium of exchange, then they will compete with fiat money (eg. US Dollar). However, if cryptocurrencies are used mainly for ‘investment purposes, hence they will compete with other classes of assets (eg. stocks or bonds)
- Baur et al (2018, p. 178) question whether Bitcoin is used as a medium of exchange or a speculative assets and they conclude that ‘Bitcoin is mainly used as a speculative investment despite or due to its high volatility and large return’. They also show that Bitcoin returns are uncorrelated with all major asset classes in normal and crises periods.

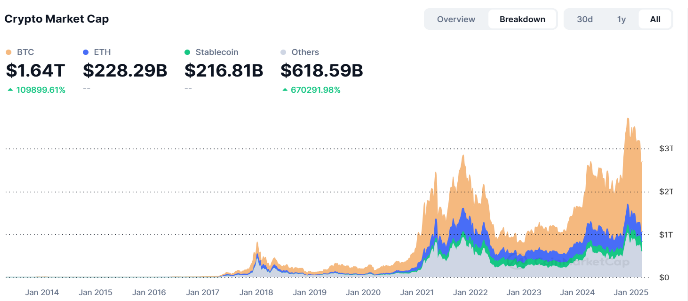
What is crypto? (2)

Ammous (2018) analyses five cryptocurrencies to evaluate whether they can fulfill the functions of money. The paper concludes that due to the inflexible supply and volatile demand, cryptocurrencies are unstable to be used as a unit of account; Only Bitcoin has the potential ‘to serve as a store of value’.

Money needs to be:

- Portable (easy to carry/move)
- Stable (in terms of value)
- Durable (not easily destroyed)
- Fungible (exchanged for similar)
- Recognizable (know it when you see it).





https://coinmarketcap.com/charts/

Bitcoin today

Review

Key Points:

- Forex volatility creates very real problems for firms
- A correlation exists between volatility and speculation
- The Tobin tax is a proposal to reduce speculation

Activity

Summarize what you learned today.

- Do you understand all the quiz answers now?
- Was this session worth your time?
- What helped you learn the most?
- What did you learn about how you learn?

References

- Ammous., 2018. Can cryptocurrencies fulfil the functions of money. The Quarterly Review of Economics and Finance. Volume 70, 38-51.
- Baur, D., Hong, K., Lee, A.D., 2018. Bitcoin: medium of exchange or speculative assets?. Journal of International Financial Markets Institutions and Money. 54, 177–189.
- Grobys, K., Junittila, J., March 2021. Speculation and Lottery-like demand in cryptocurrency markets. Journal of International Financial Markets, Institutions and Money. Volume 71
- Nakamoto, S., 2008, Bitcoin: A peer-to-peer electronic cash system, Unpublished manuscript, retrieved from https://Bitcoin.org/Bitcoin.pdf.